

Project Overview:

The Pet Adoption Center Database is outlined to streamline and oversee all viewpoints of a pet adoption center's operations. It focuses on following animals accessible for adoption, potential adopters, and the entire adoption process. The framework will store detailed data around creatures, counting their species, breed, age, medical history, and appropriation status. Furthermore, it'll keep up records of potential adopters, their inclinations, and application advance.

This database system guarantees productive organization and openness of information to disentangle workflows, upgrade creature care administration, and move forward selection results. By streamlining operations, it supports staff in coordinating pets with the correct adopters and encourages smoother selection preparation. Eventually, it points to upgrade the general involvement for both protecting staff and potential pet owners whereas guaranteeing superior results for animals in need.

Tables that we will use on this project:

1. **Animals:** Captures details like species, breed, age, gender, and adoption status to manage the shelter's animal inventory.
2. **Medical_Records:** Stores treatment dates, types, and veterinarian details to track each animal's health history.
3. **Vaccinations:** Logs vaccination information, including vaccine names and administration dates, ensuring animals are up to date on shots.
4. **Adopters:** Contains potential adopters' contact details and housing information to assess their suitability for adoption.
5. **Adoption_Applications:** Manages the application process with fields for application status and decision notes.
6. **Adoptions:** Records finalized adoptions with details about fees and staff involved in the process.
7. **Staff:** Maintains records of staff members handling various operations within the shelter.

Tables: 1. Animals:

Column Name	Data Type	Size	Null / NOT Null	Constraint	PK	Purpose of the column
animal_id	INT	-	NOT NULL	AUTO_INCREMENT	Yes	Unique identifier for each animal
name	VARCHAR	50	NOT NULL	-	No	Name given to the animal
species	VARCHAR	50	NOT NULL	-	No	Species of the animal (e.g., Dog, Cat, Rabbit)
breed	VARCHAR	50	NULL	-	No	Breed of the animal
age	INT	-	NULL	CHECK (age >= 0)		Estimated age of the animal in years
gender	ENUM	-	NOT NULL	('Male', 'Female', 'Unknown')	No	Gender of the animal
color	VARCHAR	50	NULL	-	No	Primary color of the animal
weight	DECIMAL	(5,2)	NULL	CHECK (weight > 0)	No	Weight of the animal in pounds
intake_date	DATE	-	NOT NULL	-	No	Date when the

						animal arrived at the shelter
adoption_status	ENUM	-	NOT NULL	('Available', 'Pending', 'Adopted')	No	Current adoption status of the animal

2. Medical_Records

Column Name	Data Type	Size	Null / NOT Null	Constraint	PK	Purpose of the column
record_id	INT	-	NOT NULL	AUTO_INCREMENT	Yes	Unique identifier for each medical record
animal_id	INT	-	NOT NULL	FOREIGN KEY	No	Reference to the animal
treatment_date	DATE	-	NOT NULL	-	No	Date of the medical treatment or examination
treatment_type	VARCHAR	100	NOT NULL	-	No	Type of treatment or examination performed
description	TEXT	-	NULL	-	No	Detailed description of the treatment or examination
vet_name	VARCHAR	100	NOT NULL	-	No	Name of the veterinarian who performed the treatment

3. Adopters

Column Name	Data Type	Size	Null / NOT Null	Constraint	PK	Purpose of the column
adopter_id	INT	-	NOT NULL	AUTO_INCREMENT	Yes	Unique identifier for each potential adopter
first_name	VARCHAR	50	NOT NULL	-	No	First name of the potential adopter
last_name	VARCHAR	50	NOT NULL	-	No	Last name of the potential adopter
email	VARCHAR	100	NOT NULL	UNIQUE	No	Email address of the potential adopter
phone	VARCHAR	20	NOT NULL	-	No	Phone number of the potential adopter
address	VARCHAR	255	NOT NULL	-	No	Home address of the potential adopter
housing_type	ENUM	-	NOT NULL	('House', 'Apartment', 'Other')	No	Type of housing the potential adopter lives in
has_yard	BOOLEAN	-	NOT NULL	-	No	Indicates if the potential adopter has a yard
other_pets	TEXT	-	NULL	-	No	Description of other pets owned

						by the potential adopter
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4.Vaccinations

Column Name	Data Type	Size	Null / NOT Null	Constraint	PK	Purpose of the column
vaccination_id	INT	-	NOT NULL	AUTO_INCREMENT	Yes	Unique identifier for each vaccination record
Animal_id	INT	-	NOT NULL	FOREIGN KEY	No	Reference to the animal
vaccine_name	VARCHAR	100	NOT NULL	-	No	Name of the vaccine administered
date_given	DATE	-	NOT NULL	-	No	Date when the vaccine was administered
expiration_date	DATE	-	NULL	-	No	Expiration date of the vaccine
administered_by	VARCHAR	100	NOT NULL	-	No	Name of the person who administered the vaccine

5.Adoption_Applications

Column Name	Data Type	Size	Null / NOT Null	Constraint	PK	Purpose of the column
application_id	INT	-	NOT NULL	AUTO_INCREMENT	Yes	Unique identifier for each

						adoption application
adopter_id	INT	-	NOT NULL	FOREIGN KEY	No	Reference to the potential adopter
animal_id	INT	-	NOT NULL	FOREIGN KEY	No	Reference to the animal being applied for
application_date	DATE	-	NOT NULL	-	No	Date when the application was submitted
status	ENUM	-	NOT NULL	('Pending', 'Approved', 'Rejected')	No	Current status of the application
decision_date	DATE	-	NULL	-	No	Date when a decision was made on the application
notes	TEXT	-	NULL	-	No	Additional notes or comments on the application

6.Adoptions (Fact table for completed adoptions)

Column Name	Data Type	Size	Null / NOT Null	Constraint	PK	Purpose of the column
adoption_id	INT	-	NOT NULL	AUTO_INCREMENT	Yes	Unique identifier for each completed adoption
animal_id	INT	-	NOT NULL	FOREIGN KEY	No	Reference to the adopted animal
adopter_id	INT	-	NOT NULL	FOREIGN KEY	No	Reference to the adopter
adoption_date	DATE	-	NOT NULL	-	No	Date when the adoption was finalized
adoption_fee	DECIMAL	(8,2)	NOT NULL	CHECK (adoption_fee >= 0)	No	Fee paid for the adoption
staff_id	INT	-	NOT NULL	FOREIGN KEY	No	Reference to the staff member who processed the adoption

7.Staff

Column Name	Data Type	Size	Null / NOT Null	Constraint	PK	Purpose of the column
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staff_id	INT	-	NOT NULL	AUTO_INCREMENT	Yes	Unique identifier for each staff member
first_name	VARCHAR	50	NOT NULL	-	No	First name of the staff member
last_name	VARCHAR	50	NOT NULL	-	No	Last name of the staff member
email	VARCHAR	100	NOT NULL	UNIQUE	No	Email address of the staff member
phone	VARCHAR	20	NOT NULL	-	No	Phone number of the staff member
position	VARCHAR	50	NOT NULL	-	No	Job position of the staff member
hire_date	DATE	-	NOT NULL	-	No	Date when the staff member was hired

Here **Adoptions** table will serve as a fact table because it records transactional data related to the adoption process. It captures essential details such as the animal and adopter IDs, adoption date, fees, and staff involved. This table provides a comprehensive history of finalized adoptions, enabling analysis of adoption trends and facilitating efficient management of shelter operations.